Rajiv Ranganath

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Career Overview

Final-year Computer Science student (CGPA 9.36, graduating May 2027) who builds and tests backend services: REST and WebSocket APIs in FastAPI, PostgreSQL with SQLAlchemy and Alembic, JWT auth, Docker, and pytest suites on GitHub Actions CI. AI Intern at IIT Kharagpur, building a tested LLM data pipeline. Available to start immediately, full-time or as an intern.

Education

B.N.M. Institute of Technology 2023 – 05/2027
B.E. Computer Science & Engineering Bangalore, Karnataka
CGPA: 9.36

Sri Kumarans PU College 2021 – 2023
Class XII (PCM), State Board · 92.3% Bangalore, Karnataka

Experience

AI Intern · IIT Kharagpur · Jul 2026 to Present · Remote
- Built a Python pipeline that extracts figures from company sustainability reports (PDF tables via Camelot) and certifies a value only when its source checks out; certified values match XBRL filings **97.5% (dev) and 93.3 to 100% on three of four locked holdouts**.
- Designed automatic failover across **five LLM provider APIs** (Groq, Gemini, NVIDIA NIM, OmniRoute, Ollama) with run-time model discovery, guarded by **337 pytest tests on GitHub Actions CI**.
- Audited the inherited evaluation code, removed **fabricated data and answer-key leakage**, and rebuilt scoring so every answer must cite its source.

Deep Learning Intern · E2E Management Services · Jun 2025 · Bangalore
- Built a diabetic retinopathy screening model by fine-tuning **ResNet-152** on the APTOS 2019 retina dataset to grade fundus images into five severity levels.
- Built a Streamlit app that summarises legal documents with an LLM (Groq), **flags GDPR risks** and key clauses and answers questions over the document.

Projects

AgencyDesk: Multi-Tenant Project Management Platform
- REST API in **FastAPI** with SQLAlchemy, Alembic migrations and PostgreSQL across **10 routers**, from auth and tasks to file uploads, time tracking and notifications.
- Enforced **tenant isolation in the schema** (non-nullable agency_id on every tenant table) and split global login from per-agency roles, so one user can be a client of two agencies; pytest tests prove the isolation and client-visibility rules.
- Stack: Python · FastAPI · SQLAlchemy · Alembic · PostgreSQL · JWT (bcrypt) · React · pytest

LedgerGuard: Self-Verifying Payment Reconciliation (Razorpay AI Buildathon)
- Deterministic engine reconciles orders, payments, refunds, settlements and bank records in **integer paise** against six invariants, resolving **54 of 85** held-out cases with no AI involvement.
- An LLM investigates only the cases code cannot prove, and an Evidence Gate re-checks every claim against the records, raising disposition accuracy from **89.4% to 97.7%** with **zero false auto-resolutions**.
- Stack: Python · FastAPI · Pydantic · httpx · pytest (59 tests)

Agent Console: Fault-Tolerant Real-Time WebSocket Client
- Next.js client that streams an AI agent's tokens and tool calls; a protocol engine outside React uses a **sequence buffer** to order and de-duplicate frames, and **reconnects with backoff (0.5 to 10 s)**, resuming from the last rendered message.
- Built to survive the server's **chaos mode** (mid-stream drops, reordered delivery, corrupt heartbeats); **11 Vitest tests** cover ordering, gaps, duplicates, reversed delivery and resume.
- Stack: TypeScript · Next.js · React · WebSocket · Vitest · Docker

Orbit Guard AI: Real-Time Space Debris Tracking & Collision Warning System
Tracks space debris from sparse ground-station observations and warns of likely collisions.
- Python backend (**200-particle Ensemble Kalman Filter**, GNN and JPDA observation matching) streams live tracks and collision alerts over a **FastAPI WebSocket** to a React + Three.js 3D client.
- Integrator verified against two-body physics (~10^{-12} energy drift over 24 h); a seeded 100-scenario benchmark put JPDA **6.7 points above GNN** in 8-object clutter, though half its matches were clutter.
- Stack: Python · NumPy · FastAPI · WebSocket · React · Three.js · Live demo

Skills

- Languages: Python, Java, SQL, TypeScript, JavaScript, C
- Databases & Cloud: PostgreSQL, MongoDB, AWS basics (EC2, S3, Lambda, IAM)
- Core CS (coursework): Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
- Backend: FastAPI, REST APIs, WebSocket, SQLAlchemy, Alembic, Pydantic, JWT auth
- DevOps & Testing: Docker, Git, GitHub Actions, Jenkins, pytest, Vitest
- Frontend & AI: React, Next.js, Three.js; LLM APIs (Groq, Gemini), RAG, LangChain, PyTorch

Certifications

- Introduction to Internet of Things, NPTEL (IIT Kharagpur) · Elite + Gold, top 5% · 93%